

The Notebooks · Era VII — Space

Muted purple wireframe journal

Testing Mode · UK · IB · US

Exercises

T.VII.1.001 · Column vectors GCSE notation · core · UK_GCSE_H

- (a) [never/neutral/1mk] $a=(6, 1)$, $b=(3, 2)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(6, 1)$?
- (c) [never/UK/2mk] Position $A=(6,1)$, $B=(9,3)$. Vector AB ?

T.VII.1.002 · Vector geometry: midpoint and section formula · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(5, 4)$, $b=(1, 5)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(5, 4)$?
- (c) [never/UK/2mk] Position $A=(5,4)$, $B=(6,9)$. Vector AB ?

T.VII.1.003 · Vector proof: parallelogram / midpoint theorem · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] $a=(5, 3)$, $b=(0, 2)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(5, 3)$?
- (c) [never/UK/2mk] Position $A=(5,3)$, $B=(5,5)$. Vector AB ?

T.VII.1.004 · Vector proof: collinearity and ratio · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(2, 5)$, $b=(1, 2)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(2, 5)$?
- (c) [never/UK/2mk] Position $A=(2,5)$, $B=(3,7)$. Vector AB ?

T.VII.1.005 · Position vectors in plane exam pack · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(3, 1)$, $b=(3, 4)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(3, 1)$?
- (c) [never/UK/2mk] Position $A=(3,1)$, $B=(6,5)$. Vector AB ?

T.VII.1.006 · Magnitude and unit vectors GCSE/IB · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(1, -3)$, $b=(-2, 6)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(1, -3)$?
- (c) [never/UK/2mk] Position $A=(1,-3)$, $B=(-1,3)$. Vector AB ?

T.VII.1.007 · Vector geometric proof exercise 1 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] $a=(4, 3)$, $b=(-1, 6)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(4, 3)$?
- (c) [never/UK/2mk] Position $A=(4,3)$, $B=(3,9)$. Vector AB ?

T.VII.1.008 · Vector geometric proof exercise 2 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] $a=(1, 2)$, $b=(0, 4)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(1, 2)$?
- (c) [never/UK/2mk] Position $A=(1,2)$, $B=(1,6)$. Vector AB ?

T.VII.1.009 · Vector geometric proof exercise 3 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] $a=(1, 2)$, $b=(2, 3)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(1, 2)$?
- (c) [never/UK/2mk] Position $A=(1,2)$, $B=(3,5)$. Vector AB ?

T.VII.3.001 · IB HL matrices for 2D transformations pack · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det([[1,3],[5,3]])$?
- (b) [never/IB/1mk] $[[1,3],[5,3]] \times \blacksquare$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «IB HL matrices for 2D transformations pack»: build the figure, measure, verify the claimed relation.

T.VII.3.002 · Solve 2×2 systems with inverse matrices · core · IB_AA_HL,IB_AI_HL,UK_GCSE_H

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 1 \\ 5 & 4 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 4 & 1 \\ 5 & 4 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 4 & 1 \\ 5 & 4 \end{bmatrix}$?

T.VII.3.003 · Determinant 2×2 geometric area link · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 3 & 3 \\ 5 & 2 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 3 & 3 \\ 5 & 2 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 3 & 3 \\ 5 & 2 \end{bmatrix}$?

T.VII.3.004 · Matrix multiplication non-commutativity demos · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}$?

T.VII.3.005 · AP Precalc / (+) matrices systems bridge · core · US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}\right)$?
- (b) [never/US/1mk] $\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/US/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}$?

T.VII.3.006 · Inverse of 2×2 formula fluency · core · IB_AA_HL,IB_AI_HL,UK_GCSE_H

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 4 \\ 2 & 5 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 4 & 4 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 4 & 4 \\ 2 & 5 \end{bmatrix}$?

T.VII.3.007 · Identity and zero matrix roles · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix}$?

T.VII.3.008 · Transpose properties (A+B)^T etc. · core · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix}$?

T.VII.3.009 · Solving matrix equations AX=B · challenge · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$?

T.VII.3.010 · Coded messages / Hill cipher light (fun) · strange · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}$?

T.VII.3.011 · US Precalculus matrix applications pack · core · US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 2 & 4 \\ 1 & 4 \end{bmatrix}\right)$?
- (b) [never/US/1mk] $\begin{bmatrix} 2 & 4 \\ 1 & 4 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/US/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 2 & 4 \\ 1 & 4 \end{bmatrix}$?

T.VII.3.012 · Exam mixed 2×2 matrix mega-set · challenge · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 5 \\ 2 & 5 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 4 & 5 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 4 & 5 \\ 2 & 5 \end{bmatrix}$?

T.VII.3.013 · Matrix encoding of simultaneous equations · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 5 \\ 4 & 4 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 4 & 5 \\ 4 & 4 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 4 & 5 \\ 4 & 4 \end{bmatrix}$?

T.VII.3.014 · Matrix representation of complex multiply light · strange · IB_AA_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 3 & 5 \\ 3 & 4 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 3 & 5 \\ 3 & 4 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 3 & 5 \\ 3 & 4 \end{bmatrix}$?

T.VII.5.001 · Geometric linear maps: stretch shear rotate · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 2 & 1 \\ 5 & 5 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 2 & 1 \\ 5 & 5 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Geometric linear maps: stretch shear rotate»: build the figure, measure, verify the claimed relation.

T.VII.5.002 · Composition of matrix transformations order · challenge · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 1 & 5 \\ 2 & 5 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 1 & 5 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 1 & 5 \\ 2 & 5 \end{bmatrix}$?

T.VII.5.003 · Kernel and image geometric meaning (light) · strange · IB_AA_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 3 & 5 \\ 2 & 5 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 3 & 5 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 3 & 5 \\ 2 & 5 \end{bmatrix}$?

T.VII.5.004 · Reflection matrices in x/y axes and $y=x$ · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Reflection matrices in x/y axes and $y=x$ »: build the figure, measure, verify the claimed relation.

T.VII.5.005 · Rotation matrices by θ standard form · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Write 85000 in standard form.
- (b) [never/IB/1mk] Write 4.1×10^4 as an ordinary number.
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Rotation matrices by θ standard form»: build the figure, measure, verify the claimed relation.

T.VII.5.006 · Enlargement matrix $k\mathbf{I}$ · core · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Enlargement matrix $k\mathbf{I}$ »: build the figure, measure, verify the claimed relation.

T.VII.5.007 · Shear matrices and invariant lines · challenge · IB_AI_HL,IB_AA_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 5 & 5 \\ 4 & 2 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 5 & 5 \\ 4 & 2 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Shear matrices and invariant lines»: build the figure, measure, verify the claimed relation.

T.VII.5.008 · Inverse transformation matrix meaning · core · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix}$?

T.VII.5.009 · Determinant as oriented area scale factor · challenge · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Determinant as oriented area scale factor»: build the figure, measure, verify the claimed relation.

T.VII.1.010 · N-VM (+) vectors applications bridge · core · US_CCSS_HS

- (a) [never/neutral/1mk] $a=(6, -1)$, $b=(3, 5)$. Find $a+b$ as a column vector.
- (b) [never/US/2mk] $|a|$ for $a=(6, -1)$?
- (c) [never/US/2mk] Position $A=(6,-1)$, $B=(9,4)$. Vector AB ?

T.VII.1.011 · Dot product: perpendicular test exam · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(4, -2)$, $b=(1, 4)$. Find $a+b$ as a column vector.
(b) [never/UK/2mk] $|a|$ for $a=(4, -2)$?
(c) [never/UK/2mk] Position $A=(4,-2)$, $B=(5,2)$. Vector AB ?

T.VII.1.012 · Vector equation of a line (IB) · core · IB_AA_SL,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] $a=(6, 4)$, $b=(4, 6)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(6, 4)$?
(c) [never/IB/2mk] Position $A=(6,4)$, $B=(10,10)$. Vector AB ?

T.VII.1.013 · Angle between vectors exam · core · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/1mk] $a=(5, 0)$, $b=(1, 3)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(5, 0)$?
(c) [never/IB/2mk] Position $A=(5,0)$, $B=(6,3)$. Vector AB ?

T.VII.1.014 · IB AI/AA vector applications in geometry · challenge · IB_AA_SL,IB_AA_HL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(6, 2)$, $b=(-2, 4)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(6, 2)$?
(c) [never/IB/2mk] Position $A=(6,2)$, $B=(4,6)$. Vector AB ?

T.VII.1.015 · Unit vector in direction of motion · core · IB_AA_SL,IB_AA_HL,AP_CALC_BC

- (a) [never/neutral/1mk] $a=(5, 0)$, $b=(4, 3)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(5, 0)$?
(c) [never/IB/2mk] Position $A=(5,0)$, $B=(9,3)$. Vector AB ?

T.VII.1.016 · Scalar projection applications · core · IB_AA_SL,IB_AA_HL,AP_CALC_BC

- (a) [never/neutral/1mk] $a=(5, -1)$, $b=(0, 1)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(5, -1)$?
(c) [never/IB/2mk] Position $A=(5,-1)$, $B=(5,0)$. Vector AB ?

T.VII.1.017 · 3D coordinates distance and midpoint · core · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/1mk] $a=(5, -1)$, $b=(4, 4)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(5, -1)$?
(c) [never/IB/2mk] Position $A=(5,-1)$, $B=(9,3)$. Vector AB ?

T.VII.1.018 · Vector proof of median concurrency light · strange · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(6, 2)$, $b=(3, 3)$. Find $a+b$ as a column vector.
(b) [never/UK/2mk] $|a|$ for $a=(6, 2)$?
(c) [never/UK/2mk] Position $A=(6,2)$, $B=(9,5)$. Vector AB ?

T.VII.1.019 · Displacement vs position vector clarity · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(6, 2)$, $b=(4, 3)$. Find $a+b$ as a column vector.
(b) [never/UK/2mk] $|a|$ for $a=(6, 2)$?
(c) [never/UK/2mk] Position $A=(6,2)$, $B=(10,5)$. Vector AB ?

T.VII.1.020 · IB AA HL vector product magnitude area · challenge · IB_AA_HL

- (a) [never/neutral/1mk] $a=(2, -1)$, $b=(3, 2)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(2, -1)$?
(c) [never/IB/2mk] Position $A=(2,-1)$, $B=(5,1)$. Vector AB ?

T.VII.1.021 · Resultant forces as vector sums · core · IB_AI_SL,IB_AI_HL,UK_GCSE_H

- (a) [never/neutral/1mk] $a=(6, 5)$, $b=(1, 2)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(6, 5)$?
(c) [never/IB/2mk] Position $A=(6,5)$, $B=(7,7)$. Vector AB ?

T.VII.1.022 · Unit vector i,j,k notation fluency · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(3, 5)$, $b=(1, 1)$. Find $a+b$ as a column vector.
(b) [never/UK/2mk] $|a|$ for $a=(3, 5)$?
(c) [never/UK/2mk] Position $A=(3,5)$, $B=(4,6)$. Vector AB ?

T.VII.1.023 · Parallel vectors scalar multiple test · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(6, -1)$, $b=(0, 5)$. Find $a+b$ as a column vector.
(b) [never/UK/2mk] $|a|$ for $a=(6, -1)$?
(c) [never/UK/2mk] Position $A=(6,-1)$, $B=(6,4)$. Vector AB ?

T.VII.1.024 · Vector journeys multi-stage collinearity · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] $a=(1, 5)$, $b=(1, 1)$. Find $a+b$ as a column vector.
(b) [never/UK/2mk] $|a|$ for $a=(1, 5)$?
(c) [never/UK/2mk] Position $A=(1,5)$, $B=(2,6)$. Vector AB ?

T.VII.1.025 · Component form from magnitude and direction · core · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/1mk] $a=(6, -2)$, $b=(-2, 4)$. Find $a+b$ as a column vector.
(b) [never/IB/2mk] $|a|$ for $a=(6, -2)$?
(c) [never/IB/2mk] Position $A=(6,-2)$, $B=(4,2)$. Vector AB ?

T.VII.2.001 · Gaussian elimination exam discipline · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det([[5,2],[2,3]])$?
(b) [never/IB/1mk] $[[5,2],[2,3]] \times \blacksquare$?
(c) [never/IB/3mk] If $\det \neq 0$, inverse of $[[5,2],[2,3]]$?

T.VII.2.002 · Consistent vs inconsistent systems language · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det([[4,5],[1,4]])$?
(b) [never/IB/1mk] $[[4,5],[1,4]] \times \blacksquare$?
(c) [never/IB/3mk] If $\det \neq 0$, inverse of $[[4,5],[1,4]]$?

T.VII.2.003 · Augmented matrix notation fluency · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det([[4,2],[4,5]])$?
(b) [never/IB/1mk] $[[4,2],[4,5]] \times \blacksquare$?
(c) [never/IB/3mk] If $\det \neq 0$, inverse of $[[4,2],[4,5]]$?

T.VII.2.004 · Parameter count for infinite solutions · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det([[5,5],[5,1]])$?
(b) [never/IB/1mk] $[[5,5],[5,1]] \times \blacksquare$?
(c) [never/IB/3mk] If $\det \neq 0$, inverse of $[[5,5],[5,1]]$?

T.VII.6.001 · Determinant expand vs row-reduce choice · core · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det([[1,2],[3,3]])$?
(b) [never/IB/1mk] $[[1,2],[3,3]] \times \blacksquare$?
(c) [never/IB/3mk] If $\det \neq 0$, inverse of $[[1,2],[3,3]]$?

T.VII.6.002 · Singular matrices and no inverse · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det([[3,4],[2,2]])$?
(b) [never/IB/1mk] $[[3,4],[2,2]] \times \blacksquare$?
(c) [never/IB/3mk] If $\det \neq 0$, inverse of $[[3,4],[2,2]]$?

T.VII.6.003 · Cramer's rule 2×2 (optional method) · strange · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [allowed/neutral/1mk] Notation/definition check for «Cramer's rule 2×2 (optional method)»: state the key idea in ≤ 20 words.
(b) [allowed/IB/2mk] Standard exam-style: Apply «Cramer's rule 2×2 (optional method)» in a worked example that uses the numbers 5 and 5. Show steps.
(c) [allowed/IB/3mk] IB dialect challenge: Write and solve one markable question targeting «Cramer's rule 2×2 (optional method)» (include answer).

T.VII.7.001 · Eigenvalues 2×2 exam pack (light) · challenge · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det([[1,1],[4,3]])$?
(b) [never/IB/1mk] $[[1,1],[4,3]] \times \blacksquare$?
(c) [never/IB/3mk] If $\det \neq 0$, inverse of $[[1,1],[4,3]]$?

T.VII.7.002 · Diagonalisation idea for matrix powers light · strange · IB_AA_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 4 \\ 5 & 2 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 4 & 4 \\ 5 & 2 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 4 & 4 \\ 5 & 2 \end{bmatrix}$?

T.VII.4.001 · Linear independence of 2–3 vectors in R²/R³ · core · IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $a=(5, 5)$, $b=(4, 3)$. Find $a+b$ as a column vector.
- (b) [never/IB/2mk] $|a|$ for $a=(5, 5)$?
- (c) [never/IB/2mk] Position $A=(5,5)$, $B=(9,8)$. Vector AB ?

T.VII.4.002 · Basis of R² standard and rotated · strange · IB_AA_HL

- (a) [never/neutral/1mk] $a=(5, -2)$, $b=(1, 3)$. Find $a+b$ as a column vector.
- (b) [never/IB/2mk] $|a|$ for $a=(5, -2)$?
- (c) [never/IB/2mk] Position $A=(5,-2)$, $B=(6,1)$. Vector AB ?

T.VII.8.001 · Orthogonal vectors in R² exam · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [allowed/neutral/1mk] Notation/definition check for «Orthogonal vectors in R² exam»: state the key idea in ≤ 20 words.
- (b) [allowed/IB/2mk] Standard exam-style: Apply «Orthogonal vectors in R² exam» in a worked example that uses the numbers 9 and 12. Show steps.
- (c) [allowed/IB/3mk] IB dialect challenge: Write and solve one markable question targeting «Orthogonal vectors in R² exam» (include answer).

T.VII.10.001 · Partial derivatives notation exam light · strange · IB_AA_HL,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] Notation/definition check for «Partial derivatives notation exam light»: state the key idea in ≤ 20 words.
- (b) [allowed/IB/2mk] Standard exam-style: Apply «Partial derivatives notation exam light» in a worked example that uses the numbers 10 and 11. Show steps.
- (c) [allowed/IB/3mk] IB dialect challenge: Write and solve one markable question targeting «Partial derivatives notation exam light» (include answer).

T.VII.14.001 · Vector fields intuition AP+ bridge · strange · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] $a=(4, -2)$, $b=(1, 6)$. Find $a+b$ as a column vector.
- (b) [never/US/2mk] $|a|$ for $a=(4, -2)$?
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Vector fields intuition AP+ bridge»: vary one parameter, capture before/after, write one observation.

T.VII.3.015 · Matrix transform exam figure 1 · challenge · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Matrix transform exam figure 1»: build the figure, measure, verify the claimed relation.

T.VII.1.026 · Vector geometry GCSE Higher 1 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] $a=(3, 0)$, $b=(4, 6)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(3, 0)$?
- (c) [never/UK/2mk] Position $A=(3,0)$, $B=(7,6)$. Vector AB ?

T.VII.3.016 · Matrix transform exam figure 2 · challenge · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Matrix transform exam figure 2»: build the figure, measure, verify the claimed relation.

T.VII.1.027 · Vector geometry GCSE Higher 2 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] $a=(2, 2)$, $b=(4, 5)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(2, 2)$?
- (c) [never/UK/2mk] Position $A=(2,2)$, $B=(6,7)$. Vector AB ?

T.VII.3.017 · Matrix transform exam figure 3 · challenge · IB_AA_HL,IB_AI_HL,US_CCSS_HS

(a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix}\right)$?

(b) [never/IB/1mk] $\begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix} \times I$?

(c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Matrix transform exam figure 3»: build the figure, measure, verify the claimed relation.

T.VII.1.028 · Vector geometry GCSE Higher 3 · challenge · UK_GCSE_H

(a) [never/neutral/1mk] $a=(1, 0)$, $b=(-1, 2)$. Find $a+b$ as a column vector.

(b) [never/UK/2mk] $|a|$ for $a=(1, 0)$?

(c) [never/UK/2mk] Position $A=(1,0)$, $B=(0,2)$. Vector AB ?

T.VII.3.018 · Matrix transform exam figure 4 · challenge · IB_AA_HL,IB_AI_HL,US_CCSS_HS

(a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix}\right)$?

(b) [never/IB/1mk] $\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix} \times I$?

(c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Matrix transform exam figure 4»: build the figure, measure, verify the claimed relation.

T.VII.1.029 · Vector geometry GCSE Higher 4 · challenge · UK_GCSE_H

(a) [never/neutral/1mk] $a=(2, 2)$, $b=(1, 2)$. Find $a+b$ as a column vector.

(b) [never/UK/2mk] $|a|$ for $a=(2, 2)$?

(c) [never/UK/2mk] Position $A=(2,2)$, $B=(3,4)$. Vector AB ?

T.VII.5.010 · Standard matrix recognition 1 · core · IB_AA_HL,IB_AI_HL

(a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}\right)$?

(b) [never/IB/1mk] $\begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \times I$?

(c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Standard matrix recognition 1»: build the figure, measure, verify the claimed relation.

T.VII.5.011 · Standard matrix recognition 2 · core · IB_AA_HL,IB_AI_HL

(a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 1 \\ 3 & 5 \end{bmatrix}\right)$?

(b) [never/IB/1mk] $\begin{bmatrix} 4 & 1 \\ 3 & 5 \end{bmatrix} \times I$?

(c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Standard matrix recognition 2»: build the figure, measure, verify the claimed relation.

T.VII.3.019 · Matrix transform area scale |det| interpretation · core · IB_AA_HL,IB_AI_HL,US_CCSS_HS

(a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix}\right)$?

(b) [never/IB/1mk] $\begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix} \times I$?

(c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Matrix transform area scale |det| interpretation»: build the figure, measure, verify the claimed relation.

Answer Key — Era VII

Brief solutions. Full reasoning in machine JSON.

T.VII.1.001

- (a) (9, 3) — *Add components.*
- (b) $\sqrt{37} - \sqrt{(x^2+y^2)}$.
- (c) (3, 2) — $B-A$.

T.VII.1.002

- (a) (6, 9) — *Add components.*
- (b) $\sqrt{41} - \sqrt{(x^2+y^2)}$.
- (c) (1, 5) — $B-A$.

T.VII.1.003

- (a) (5, 5) — *Add components.*
- (b) $\sqrt{34} - \sqrt{(x^2+y^2)}$.
- (c) (0, 2) — $B-A$.

T.VII.1.004

- (a) (3, 7) — *Add components.*
- (b) $\sqrt{29} - \sqrt{(x^2+y^2)}$.
- (c) (1, 2) — $B-A$.

T.VII.1.005

- (a) (6, 5) — *Add components.*
- (b) $\sqrt{10} - \sqrt{(x^2+y^2)}$.
- (c) (3, 4) — $B-A$.

T.VII.1.006

- (a) (-1, 3) — *Add components.*
- (b) $\sqrt{10} - \sqrt{(x^2+y^2)}$.
- (c) (-2, 6) — $B-A$.

T.VII.1.007

- (a) (3, 9) — *Add components.*
- (b) $5 - \sqrt{(x^2+y^2)}$.
- (c) (-1, 6) — $B-A$.

T.VII.1.008

- (a) (1, 6) — *Add components.*
- (b) $\sqrt{5} - \sqrt{(x^2+y^2)}$.
- (c) (0, 4) — $B-A$.

T.VII.1.009

- (a) (3, 5) — *Add components.*
- (b) $\sqrt{5} - \sqrt{(x^2+y^2)}$.
- (c) (2, 3) — $B-A$.

T.VII.3.001

- (a) $-12 - ad - bc$.
- (b) $[[1,3],[5,3]]$ — *Identity.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VII.3.002

- (a) $11 - ad - bc$.
- (b) $[[4,1],[5,4]]$ — *Identity.*
- (c) $(1/11)[[4,-1],[-5,4]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.3.003

- (a) $-9 - ad - bc$.
- (b) $[[3,3],[5,2]]$ — *Identity.*
- (c) $(1/-9)[[2,-3],[-5,3]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.3.004

- (a) $0 - ad - bc$.
- (b) $[[2,3],[2,3]]$ — *Identity.*
- (c) No inverse — $(1/\det)[[d,-b],[-c,a]]$.

T.VII.3.005

- (a) $-24 - ad - bc$.

- (b) $[[1,5],[5,1]] \text{ --- Identity.}$
 (c) $(1/-24)[[1,-5],[-5,1]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.006

- (a) $12 \text{ --- } ad-bc.$
 (b) $[[4,4],[2,5]] \text{ --- Identity.}$
 (c) $(1/12)[[5,-4],[-2,4]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.007

- (a) $2 \text{ --- } ad-bc.$
 (b) $[[2,4],[2,5]] \text{ --- Identity.}$
 (c) $(1/2)[[5,-4],[-2,2]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.008

- (a) $0 \text{ --- } ad-bc.$
 (b) $[[5,3],[5,3]] \text{ --- Identity.}$
 (c) No inverse $\text{--- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.009

- (a) $1 \text{ --- } ad-bc.$
 (b) $[[4,3],[1,1]] \text{ --- Identity.}$
 (c) $(1/1)[[1,-3],[-1,4]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.010

- (a) $14 \text{ --- } ad-bc.$
 (b) $[[5,3],[2,4]] \text{ --- Identity.}$
 (c) $(1/14)[[4,-3],[-2,5]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.011

- (a) $4 \text{ --- } ad-bc.$
 (b) $[[2,4],[1,4]] \text{ --- Identity.}$
 (c) $(1/4)[[4,-4],[-1,2]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.012

- (a) $10 \text{ --- } ad-bc.$
 (b) $[[4,5],[2,5]] \text{ --- Identity.}$
 (c) $(1/10)[[5,-5],[-2,4]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.013

- (a) $-4 \text{ --- } ad-bc.$
 (b) $[[4,5],[4,4]] \text{ --- Identity.}$
 (c) $(1/-4)[[4,-5],[-4,4]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.3.014

- (a) $-3 \text{ --- } ad-bc.$
 (b) $[[3,5],[3,4]] \text{ --- Identity.}$
 (c) $(1/-3)[[4,-5],[-3,3]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.5.001

- (a) $5 \text{ --- } ad-bc.$
 (b) $[[2,1],[5,5]] \text{ --- Identity.}$
 (c) Lab evidence (notes/screenshot) + one verified check $\text{--- Complete lab steps.}$

T.VII.5.002

- (a) $-5 \text{ --- } ad-bc.$
 (b) $[[1,5],[2,5]] \text{ --- Identity.}$
 (c) $(1/-5)[[5,-5],[-2,1]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.5.003

- (a) $5 \text{ --- } ad-bc.$
 (b) $[[3,5],[2,5]] \text{ --- Identity.}$
 (c) $(1/5)[[5,-5],[-2,3]] \text{ --- } (1/det)[[d,-b],[-c,a]].$

T.VII.5.004

- (a) $2 \text{ --- } ad-bc.$
 (b) $[[3,2],[2,2]] \text{ --- Identity.}$
 (c) Lab evidence (notes/screenshot) + one verified check $\text{--- Complete lab steps.}$

T.VII.5.005

- (a) $8.5 \times 10^4 \text{ --- } 1 \leq A < 10.$
 (b) 41000 $\text{--- Shift decimal.}$
 (c) Lab evidence (notes/screenshot) + one verified check $\text{--- Complete lab steps.}$

T.VII.5.006

- (a) $9 - ad - bc$.
 (b) $[[5,3],[2,3]] - \text{Identity}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.VII.5.007

- (a) $-10 - ad - bc$.
 (b) $[[5,5],[4,2]] - \text{Identity}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.VII.5.008

- (a) $0 - ad - bc$.
 (b) $[[3,4],[3,4]] - \text{Identity}$.
 (c) No inverse — $(1/\det)[[d,-b],[-c,a]]$.

T.VII.5.009

- (a) $-6 - ad - bc$.
 (b) $[[1,3],[3,3]] - \text{Identity}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.VII.1.010

- (a) $(9, 4) - \text{Add components}$.
 (b) $\sqrt{37} - \sqrt{(x^2+y^2)}$.
 (c) $(3, 5) - B-A$.

T.VII.1.011

- (a) $(5, 2) - \text{Add components}$.
 (b) $\sqrt{20} - \sqrt{(x^2+y^2)}$.
 (c) $(1, 4) - B-A$.

T.VII.1.012

- (a) $(10, 10) - \text{Add components}$.
 (b) $\sqrt{52} - \sqrt{(x^2+y^2)}$.
 (c) $(4, 6) - B-A$.

T.VII.1.013

- (a) $(6, 3) - \text{Add components}$.
 (b) $5 - \sqrt{(x^2+y^2)}$.
 (c) $(1, 3) - B-A$.

T.VII.1.014

- (a) $(4, 6) - \text{Add components}$.
 (b) $\sqrt{40} - \sqrt{(x^2+y^2)}$.
 (c) $(-2, 4) - B-A$.

T.VII.1.015

- (a) $(9, 3) - \text{Add components}$.
 (b) $5 - \sqrt{(x^2+y^2)}$.
 (c) $(4, 3) - B-A$.

T.VII.1.016

- (a) $(5, 0) - \text{Add components}$.
 (b) $\sqrt{26} - \sqrt{(x^2+y^2)}$.
 (c) $(0, 1) - B-A$.

T.VII.1.017

- (a) $(9, 3) - \text{Add components}$.
 (b) $\sqrt{26} - \sqrt{(x^2+y^2)}$.
 (c) $(4, 4) - B-A$.

T.VII.1.018

- (a) $(9, 5) - \text{Add components}$.
 (b) $\sqrt{40} - \sqrt{(x^2+y^2)}$.
 (c) $(3, 3) - B-A$.

T.VII.1.019

- (a) $(10, 5) - \text{Add components}$.
 (b) $\sqrt{40} - \sqrt{(x^2+y^2)}$.
 (c) $(4, 3) - B-A$.

T.VII.1.020

(a) (5, 1) — Add components.

(b) $\sqrt{5} - \sqrt{(x^2+y^2)}$.

(c) (3, 2) — $B-A$.

T.VII.1.021

(a) (7, 7) — Add components.

(b) $\sqrt{61} - \sqrt{(x^2+y^2)}$.

(c) (1, 2) — $B-A$.

T.VII.1.022

(a) (4, 6) — Add components.

(b) $\sqrt{34} - \sqrt{(x^2+y^2)}$.

(c) (1, 1) — $B-A$.

T.VII.1.023

(a) (6, 4) — Add components.

(b) $\sqrt{37} - \sqrt{(x^2+y^2)}$.

(c) (0, 5) — $B-A$.

T.VII.1.024

(a) (2, 6) — Add components.

(b) $\sqrt{26} - \sqrt{(x^2+y^2)}$.

(c) (1, 1) — $B-A$.

T.VII.1.025

(a) (4, 2) — Add components.

(b) $\sqrt{40} - \sqrt{(x^2+y^2)}$.

(c) (-2, 4) — $B-A$.

T.VII.2.001

(a) $11 - ad - bc$.

(b) $[[5,2],[2,3]]$ — Identity.

(c) $(1/11)[[3,-2],[-2,5]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.2.002

(a) $11 - ad - bc$.

(b) $[[4,5],[1,4]]$ — Identity.

(c) $(1/11)[[4,-5],[-1,4]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.2.003

(a) $12 - ad - bc$.

(b) $[[4,2],[4,5]]$ — Identity.

(c) $(1/12)[[5,-2],[-4,4]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.2.004

(a) $-20 - ad - bc$.

(b) $[[5,5],[5,1]]$ — Identity.

(c) $(1/-20)[[1,-5],[-5,5]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.6.001

(a) $-3 - ad - bc$.

(b) $[[1,2],[3,3]]$ — Identity.

(c) $(1/-3)[[3,-2],[-3,1]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.6.002

(a) $-2 - ad - bc$.

(b) $[[3,4],[2,2]]$ — Identity.

(c) $(1/-2)[[2,-4],[-2,3]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.6.003

(a) Accept accurate paraphrase of: Solve 2×2 via Cramer; know limitations. — On-skill definition.

(b) Correct worked application of Cramer's rule 2×2 (optional method) using 5, 5 — Use method from «Cramer's rule 2×2 (optional method)»; compute carefully with 5, 5.

(c) Student-authored on-skill Q + correct solution — Must use track vocabulary and be markable.

T.VII.7.001

(a) $-1 - ad - bc$.

(b) $[[1,1],[4,3]]$ — Identity.

(c) $(1/-1)[[3,-1],[-4,1]] - (1/\det)[[d,-b],[-c,a]]$.

T.VII.7.002

(a) $-12 - ad - bc$.

- (b) $[[4,4],[5,2]]$ — Identity.
 (c) $(1/-12)[[2,-4],[-5,4]]$ — $(1/\det)[[d,-b],[-c,a]]$.

T.VII.4.001

- (a) (9, 8) — Add components.
 (b) $\sqrt{50} - \sqrt{(x^2+y^2)}$.
 (c) (4, 3) — $B-A$.

T.VII.4.002

- (a) (6, 1) — Add components.
 (b) $\sqrt{29} - \sqrt{(x^2+y^2)}$.
 (c) (1, 3) — $B-A$.

T.VII.8.001

- (a) Accept accurate paraphrase of: Perpendicular via dot product practice. — On-skill definition.
 (b) Correct worked application of Orthogonal vectors in R^2 exam using 9, 12 — Use method from «Orthogonal vectors in R^2 exam»; compute carefully with 9, 12.
 (c) Student-authored on-skill Q + correct solution — Must use track vocabulary and be markable.

T.VII.10.001

- (a) Accept accurate paraphrase of: Compute simple f_x, f_y for IB/AP+ curiosity. — On-skill definition.
 (b) Correct worked application of Partial derivatives notation exam light using 10, 11 — Use method from «Partial derivatives notation exam light»; compute carefully with 10, 11.
 (c) Student-authored on-skill Q + correct solution — Must use track vocabulary and be markable.

T.VII.14.001

- (a) (5, 4) — Add components.
 (b) $\sqrt{20} - \sqrt{(x^2+y^2)}$.
 (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VII.3.015

- (a) -2 — $ad-bc$.
 (b) $[[2,1],[4,1]]$ — Identity.
 (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VII.1.026

- (a) (7, 6) — Add components.
 (b) $3 - \sqrt{(x^2+y^2)}$.
 (c) (4, 6) — $B-A$.

T.VII.3.016

- (a) 0 — $ad-bc$.
 (b) $[[2,4],[2,4]]$ — Identity.
 (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VII.1.027

- (a) (6, 7) — Add components.
 (b) $\sqrt{8} - \sqrt{(x^2+y^2)}$.
 (c) (4, 5) — $B-A$.

T.VII.3.017

- (a) -17 — $ad-bc$.
 (b) $[[1,4],[5,3]]$ — Identity.
 (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VII.1.028

- (a) (0, 2) — Add components.
 (b) $1 - \sqrt{(x^2+y^2)}$.
 (c) (-1, 2) — $B-A$.

T.VII.3.018

- (a) -13 — $ad-bc$.
 (b) $[[1,4],[4,3]]$ — Identity.
 (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VII.1.029

- (a) (3, 4) — Add components.
 (b) $\sqrt{8} - \sqrt{(x^2+y^2)}$.
 (c) (1, 2) — $B-A$.

T.VII.5.010

- (a) 11 — $ad-bc$.

(b) $[[3,1],[1,4]]$ — *Identity*.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.VII.5.011

(a) $17 - ad - bc$.

(b) $[[4,1],[3,5]]$ — *Identity*.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.VII.3.019

(a) $-3 - ad - bc$.

(b) $[[2,5],[1,1]]$ — *Identity*.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.